Larris Xie

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Education

University of Waterloo Apr 2028
Bachelor's of Computer Science **3.9/4.0 GPA**

Experience

Conway Apr 2026 - Aug 2026
Software Engineering Intern

- Selected as 1 of 40 **Kleiner Perkins Fellows** from 5000+ candidates
- Built a GPU **inference service for transformers** on Triton, TensorRT, and EKS, using a tiered **KV cache** with predictive prefetching to reach a **95%** cache hit rate and cut p95 latency by **8×**.
- Created an **durable agent orchestration** runtime for fraud response using the Claude Agent SDK, Python, and DBOS, executing workflows over **autoscaling K8s Jobs** at peaks of **17M events/day**.
- Architected a live **SQL rules engine** in Python and Snowflake that compiles stateful SQL into Snowflake Tasks, powering fraud detection across 1.7B+ events.

University of Waterloo Jan 2025 - May 2026
Undergraduate Research Assistant

- **Published as a co-first author** on PPHH-VFL, a **privacy preserving inference architecture** for transformer-based vertical federated learning on distributed time-series data.
- Developed custom **PyTorch** transformer models with adversarial GAN privacy training, achieving a 95% F1 score on fraud detection under a 10,000:1 fraud-to-legitimate transaction ratio.
- Accelerated **private transformer inference** by **up to 44.4×** while retaining 100% of performance by engineering a distributed GPU framework with CrypTen MPC and PyTorch RPC.

Shopify May 2025 - Aug 2025
Software Engineering Intern

- Orchestrated a **100% uptime data migration** from legacy APIs to new service endpoints while ensuring full **backwards compatibility** through shadow writes, backfilling **over 5B rows**, and staged rollouts.
- Designed and implemented scalable **data models and GraphQL APIs** in Ruby on Rails, powering a fully extensible cash management experience for **2M+ active retailers**.
- Engineered full-stack features for Shopify Point of Sale (POS) in React Native and Typescript, resolving critical race conditions in async workflows to reduce failure rate by **40%.**

Meta x MLH May 2025 - Aug 2025
Production Engineering Fellow

- Deployed and managed **Linux infrastructure** on CentOS, leveraging system calls and kernel-level debugging at the OS layer, and Docker and NGINX at the application layer to improve reliability.
- Configured system monitoring with Prometheus, cAdvisor, and Grafana, leveraging low-level resource metrics (CPU, cache misses, memory usage, I/O throughput) to cut bug detection time by **80%.**
- Automated **CI/CD workflows** with GitHub Actions, integrating testing frameworks, security scans, and performance monitoring to streamline deployments and improve system reliability.

Publications

Fenaux, Lucas*, **Larris Xie***, Aditya Bang, et al. (2026). "Private Vertical Federated Inference for Time-Series." *arXiv preprint arXiv:2605.08343.* Under submission at **ICLR 2027**. * Equal contribution.

Projects

Self-Taught Long Context for Decision Models *PyTorch, LoRA, CUDA, Hugging Face*

- Developed a **label free self-distillation** method that trained a 0.8B decision model to handle **256k** token inputs from **12% to 75%** accuracy, also beating supervised training on real labels.
- Rewrote the model's **KV cache branch evaluation** to score queries in memory-bounded chunks off a single prefix pass, enabling 256k token inputs on one GPU at **8× speed** while retaining 100% parity.

LifeOS (Hack the 6ix Winner) *React, TypeScript, Python, FastAPI, Redis, Docker, PostgreSQL, AWS*

- Created a real-time multimodal MCP layer that continuously captures in-person experiences and turns them into searchable, actionable context for LLMs and automations.
- Engineered a **distributed AI video processing pipeline** using Redis Queue that orchestrates semantic analysis with TwelveLabs, embedding generation, and cloud storage via asynchronous workers.